

Hidden dynamics set the limit—and become the signal—in ghost imaging

[author block — USER DECISION (gap G7)]

Abstract

[ABSTRACT — draft to the 6-beat sheet above, 95–105 words. Numeric callouts limited to +1.87 dB, 19.13×, and $\leq 1.5\times$ pilot ($\langle C: \text{pilot ratio} \rangle$). No displayed equations, no internal campaign names, no DLGI acronym (R35 cut 7).]

1 One hidden channel, two consequences

A bucket camera records one number per pattern, but that number is produced by a dynamic hidden channel.^[GI/SPI core] Detector live time and medium fluctuations are usually treated as distortions to be corrected; here they determine both what the scene channel can never recover and what the same record can measure instead.

► TODO: Finish para 1 (2–3 sentences). State the two consequences immediately: the hidden state can *erase* scene information, but its temporal imprint can *identify* the medium. End the paragraph on the promise phrase “ghost imaging becomes its own medium monitor.” Source: R35 s3.1 para 1; materials map s1a intro seed (paper/main_oe.tex 72–141) + s2a (paper2/main_m1.tex 61–97). Do NOT open with “Ghost imaging has attracted...”, an application list, dead-time definitions, DCS history, or a contribution list (R35 s3.1).

► TODO: Para 2 (the boundary). State that an exact missing-information specialization turns detector-state uncertainty into a measurable scene- information loss and a jitter-capped ridge. Give ONLY the two empirical consequences: global power control and the confirmed imaging/time gains (+1.87 dB, 19.13×). Source: R35 s3.1 para 2; identity from paper2 Thm 1–2 (materials map s2a 118–167) + paper-1 Eq. (missing) (materials map s1a 241–378).

► TODO: Para 3. A positive operating law is not yet a boundary, so five preregistered closure tests probe the remaining count-only degrees of freedom; their joint negative result motivates *reusing*, not correcting away, the hidden medium state. Source: R35 s3.1 para 3. END the paragraph – and the section – with the thesis sentence (frozen, R35 s1.1, in the abstract comment block above).

We first quantify the information removed by the hidden state before asking how that boundary

should be used.

2 Hidden-state information sets a finite ridge

► TODO: Opening sentence: credit Louis (observed information in EM) and the Orchard–Woodbury missing-information lineage, THEN state the optical specialization as the contribution. Source: R35 s3.2 + s7.5; DOI fences in materials map s8c (Louis 10.1111/j.2517-6161.1982.tb01203.x).

► TODO: Eq. (1) – the observed-score conditional expectation (hidden-state identity, single form with detector *and* medium specializations; do NOT brand separate E1/O1 – R35 cut 3). Import: paper2 Thm 1 “missing-information identity” $I_N = \mathbb{E}[N_T] - \lambda^2 \mathbb{E}[\text{Var}(L_T | N_T)]$ (materials map s2a 118–167, main_m1.tex Eq. (identity)); the paper-1 terminal-phase form $I_N = \mathbb{E}[N] - \rho^2 \mathbb{E}[\text{Var}(R_\nu | N)]$ (materials map s1a Eq. (missing), main_oe.tex 253–257) is the same identity in the deterministic special case.

► TODO: Eq. (2) – the missing-information identity (long-window rate). Import: paper2 Thm 2 $J_\infty(\rho, c_v) = \rho / [(1 + \rho)(1 + c_v^2 \rho^2)]$ (main_m1.tex Eq. (Jinf)).

► TODO: Eq. (3) – one dead-time specialization exposing the hidden exposure/live-time covariance (the jitter mechanism), leading to...

► TODO: Eq. (4) – the frozen ridge law + asymptotic exponents in existing notation. Import the jitter-capped optimum paper2 Thm 3 $c_v^2 \rho_c^2 (1 + 2\rho_c) = 1$ with $\rho_c = 2^{-1/3} c_v^{-2/3} - \frac{1}{6} + \dots$ (main_m1.tex Eq. (cubic)/(smalle)), the deterministic cube-root ridge $\rho^*(\nu) = (6\nu)^{1/3} - 2/3 + O(\nu^{-1/3})$ (paper-1 Eq. (cube-root), main_oe.tex 281–284), and the crossover bridge $\rho_*(\nu, c_v) \sim (6\nu)^{1/3} (1 + 12\nu c_v^2)^{-1/3}$ (main_m1.tex Eq. (crossover)) that UNIFIES the two theorem sets into one Act-I progression (gap G8). Frozen blind-verified pooled slope -0.658 vs $-2/3$ (materials map s2b S4).

The hidden-state identity therefore yields an operating point, not an estimator.

The theorem identifies a finite operating point; the next question is whether one scalar control can realize it in imaging.

FIG. 1 (HERO) PLACEHOLDER --- limit $\rightarrow$ reach $\rightarrow$ close $\rightarrow$ reuse. Stage 1: one hidden-state bucket experiment (label chain ‘simulation model’). Stage 2: scene-info arrow to finite ridge + global dial + gray ‘count-only scene boundary’ wall w/ five unlabeled ticks; callouts +1.87 dB, 19.13 $\times$ only. Stage 3 (PASS): raw stream branches to scene card + medium card (t_c , CV) with reciprocity symbol κ_i^2 , ‘0 extra exposures’, ‘ $\leq 1.5\times$ pilot’, ≤ 4 colors.

Figure 1: **[Fig. 1 title — the 10-second paper.]** [Caption 120–160 words (R35 s2.1). What the referee must grasp in ten seconds (R35 s6.1): (1) hidden dynamics cap scene information; (2) a global dial reaches the cap and yields a large speed gain; (3) count-only adaptations do not materially move the cap; (4) the same record can instead become a medium instrument. No provenance narrative in the caption (R35 cut 13).]

FIG. 2 PLACEHOLDER --- hidden-state ridge. (a) complete vs observed bucket experiment; hidden state + missing-information subtraction, shown once. (b) jitter-capped $J(\rho; c)$ for deterministic + two jitter levels, optima marked. (c) blind verification of the two frozen scaling exponents on log--log axes (theory lines + independent simulation points).

Figure 2: **[Fig. 2 title.]** [Caption 120–160 words; the ridge is believed before any image is shown. Source: paper2 hero+validation figs (materials map s6b) reused; (a) panel is new. No provenance narrative.]

3 One global control reaches the ridge

► TODO: Beat 1 (one sentence). M1 design: fixed mask bank, safe comparator, ridge multiplier, preregistered scene cohort. Source: paper2 s4–s5 (main_m1.tex 232–292); full protocol -i Supplement S4. Translate internal labels into reader language – no SCAT32, no RIDGE_OPERATING_PASS in the main text (R35 s3.3 “must not contain”).

The preregistered operating-point endpoint passed: at $\nu = 2000$, ridge control improved median fixed-dwell radiometric quality over the matched 0.60 comparator by +1.87 dB, with a family-stratified lower bound of 1.41 dB and 19 of 24 scenes positive. The same operating point used 37.1 $\times$ the incident dose and 2.6 $\times$ the detected counts.

Against the safe reference, the median conservative speed ratio was 19.13 $\times$, the family-stratified nested-bootstrap 2.5th-percentile bound was 18.33 $\times$, and 21 of 24 scenes had $S_{\text{gate}} > 1$.

These are fixed-dwell and elapsed-time benefits bought with power, not photon-efficiency gains.

► TODO: Beat 4 (one sentence). Edge/failure cases: median family gain ranges +6.37 dB (maze) to –4.18 dB (contour); per-scene gain correlates with bucket contrast at +0.64; no guard explains the negatives – they delimit where one global multiplier is insufficiently aligned with scene structure. Source: paper2 s6 (main_m1.tex 372–380). Full per-family/per-scene tables -i S5.

► TODO: Beat 5 (one sentence). Point to the post-unblinding analysis correction and immutable provenance in Methods / Supplement S4; do NOT narrate it here. Frozen correction record: register s9a (three discrepancies; “agreed on all 18 audited numerical outputs”); source-of-record [results/round63_m1/CORRECTION_2026-07-19/](#).

The global dial works; the remaining question is whether richer control of the same scalar count can materially improve it.

A global dial reaches the ridge, but that does not yet show that richer count-only design or inference cannot materially surpass it.

FIG. 3 PLACEHOLDER --- ridge realized in imaging. Left third: two large reconstruction comparisons (one representative positive, one boundary case) -- NO gallery. Upper right: 24-scene fixed-dwell ΔQ forest plot (median, LB, zero line). Lower right: elapsed-time speed distribution with the 3 $\times$ gate, median 19.13 $\times$, LB 18.33 $\times$, censoring symbols; optional small quality-vs-time curve.

Figure 3: **[Fig. 3 title.]** [Caption 120–160 words. Translate **FAST_RIGHT_CENSORED** etc. into reader language (R35 s3.3). No audit hashes, frozen cell IDs, solver caps, or process terminology.]

4 Closing the integrated-count scene channel 5 One record, two scientific outputs

The scientific question is singular: is the global ridge merely a convenient baseline, or is it the practical optimum of the declared count-only experiment? To answer it, five preregistered interventions share one common comparator, one common resource ledger, and a pre-registered 1 dB materiality bar on image quality, and they divide into the two families of freedom that remain after the detector has compressed each exposure to a single count: design-side freedoms (the illumination geometry, the measurement allocation, and the temporal order) and inference-side freedoms (the estimator and the count likelihood).

On the design side, none of the three interventions clears the bar: scene-adapted geometry drawn from a dose-uniform library reaches only +0.68 dB over the ridge baseline, robust reallocation of the measurement budget is actively harmful at −8.3 dB, and optimizing the temporal order of the same patterns yields +0.04 dB.

On the inference side, the renewal quasi-likelihood estimator is already within roughly one per cent of the certified count-Fisher ceiling, leaving at most 0.2 dB of headroom, and replacing it with the exact conditional-count likelihood returns +0.41 dB. Together these tests do not prove a universal optimum over every imaginable optical measurement; they close the declared integrated-count class from both the design and software sides. Once the scalar scene channel is closed, the remaining question is not how to manufacture another fraction of a decibel from the same count, but whether the hidden channel can become a second output.

► TODO: Single-claim sentence: when the hidden medium state carries parameters of scientific interest, retaining the raw bucket stream allows joint scene+medium inference; the same canonical confusion modes tax both ledgers, and nuisance-orthogonal scheduling can improve both rather than trading one against the other. Source: R35 s5.1.

$$\mathcal{I} = \begin{bmatrix} A & C \\ C^\top & B \end{bmatrix}. \quad (1)$$

► TODO: Define the two Schur-complement ledgers and the whitened cross operator $K = A^{-1/2}CB^{-1/2}$. State (without long proof) that $KK^\top$ and $K^\top K$ share the same nonzero eigenvalues κ_i^2 , giving equal normalized determinant retention $\det(J_x)/\det(A) = \prod(1 - \kappa_i^2) = \det(J_\theta)/\det(B)$ – the Canonical-Confusion Ledger Reciprocity result. Import: R33 Thm 1–2 (materials map s4b). Full proof, pseudoinverse version, path-vs-hyperparameter distinction, and the (A, B, K) matrices -i Supplement S7.

► TODO: State the O4-A consequence plainly (R35 s5.2, one sentence): “Reducing the cross block while preserving the oracle ledgers protects both products; paired scheduling is not a transfer of information from scene to medium.” FORBIDDEN wording (register s9c / R33 s4c): no “information conservation”, no “free medium information”, no “paired schedules trade scene into medium”, no additive conservation law. Correct framing: “two projections of the same posterior ambiguity, coupled through the posterior cross-covariance K ”.

Dedicated diffusing-wave and diffuse-correlation spectroscopies already estimate medium dynamics, but they do so from purpose-built temporal intensity fluctuations rather than from an imaging record.^[DWS/DCS] Pilot-interleaved and interpolated ghost-imaging monitors already insert dedicated reference measurements to track medium gain, at the cost of the frames those references consume.^[pilot GI] The distinction claimed here is the simultaneous recovery of an unknown scene and

FIG. 4 PLACEHOLDER --- count-only closure. (a) intervention map: pattern-design $\rightarrow$ count-likelihood pipeline with five intervention points, all enclosed in one box “same integrated-count observation”. (b) common materiality plot: incremental dB over the ridge baseline on one axis with a shaded 1 dB region; design-side brace $\{+0.68, -8.3, +0.04\}$, inference-side brace $\{\leq 0.2 \text{ headroom (capped arrow), } +0.41\}$; broken-axis inset for -8.3 dB . Right edge: one arrow leaving the box SIDEWAYS labeled “second scientific output”.

Figure 4: **[Fig. 4 title: Double-sided closure of the count-only scene channel.]** [Caption 120–160 words. Teach that the campaigns are orthogonal probes of ONE class, not a chronology. Internal acronyms + full campaign tables live in Supplement S6, nowhere else (R35 s4.3).]

quantitative medium hyperparameters from the same ordinary programmable bucket record — with no pilot or reference acquisition, and with a joint scene/medium confusion certificate.

- TODO: Install the R34 authorized claim (frozen, register s9d, verbatim): “DLGI extracts a competitive point estimate of medium correlation time and fluctuation strength from the same bucket record used for scene reconstruction, with no added acquisition and no detected scene penalty; model-calibrated t_c uncertainty remains under confirmation.” On C1-C7 PASS, restore the R33 headline “One bucket stream, two model-certified products...” with “model-certified” + validated scalar-gain domain MANDATORY (register s9d).
- TODO: Beat 1 – byte-identical / equal-ledger statement (identical photons, exposures, multiset; only order+inference change). (C: identical-ledger confirm).
- TODO: Beat 2 – model/gauge residual check in one sentence (scalar-gain model + gauge pass held-out residual tests). (C: innovation std, lag-1, gauge shift).
- TODO: Beat 3 – pilot-free vs pilot and oracle precision for t_c and CV. (C: RMSE ratios $\leq 1.5 \times \text{pilot}$).
- TODO: Beat 4 – calibrated interval coverage and width. (C: coverage $\in [0.92, 0.98]$ per cell; width $\leq 1.5 \times \text{pilot}$). (This is C1/C2, the bar that FAILED in the feasibility probe – materials map s4a bar 4; the whole section is contingent on it clearing in the frozen-Neyman campaign.)
- TODO: Beat 5 – scene noninferiority. (C: $\Delta \text{PSNR} \geq -0.2 \text{ dB}$ vs byte-identical linear comparator).
- TODO: Beat 6 – schedule ordering predicted by (A, B, K) . (C: reciprocity det match; paired best for both ledgers).
- TODO: Beat 7 – one compact edge-domain statement (fast-drift, slow-drift, weak-CV/noise failure regions). (C: edge map).

The same hidden state that fixes the scene-information boundary can therefore become a model-certified medium measurement without a second acquisition.

6 Boundary, scope, and experimental transfer

► TODO: Discussion para 1 – the boundary-to-resource principle. State the result as: hidden dynamics set the scene-information boundary, and the SAME hidden state supplies a second measurement once its model is validated. Refer back to Figs. 1 and 5; introduce no new result. Source: R35 s2.2 + the s5 “transition out” frozen beam.

► TODO: MOLT slot – EXACTLY ONE sentence pointing to Supplement S9 as evidence that adding a mathematically new channel is not sufficient when the channel is Fisher-weak. No MOLT acronym in the main text (R35 s4.4). Source: materials map s5a; S9 title “Detector-side escape beyond the count-only model: a real operator with insufficient matched-resource image strength.”

► TODO: Bench-falsification paragraph – specify the minimal rotating-diffuser validation: same pattern stream, one bucket detector, independently controllable decorrelation time, external monitor used ONLY as ground truth. State the exact physical violation that kills the model: spatially varying transfer rather than scalar gain (R35 s7.4). Source: paper2 “immediate bench test” (main_m1.tex 430–438) + R35 s7.1. This paragraph pre-empts “there is no experiment”.

► TODO: Three-sentence conclusion. Boundary-to-resource principle; declared scalar-gain/detector domain; the bench falsification path. Do NOT repeat the abstract (R35 cut 14).

Methods and provenance capsule

► TODO: Methods stub. Cover, tersely: (i) the exact statistical model and the “simulation model” labeling of every optical schematic (R35 s7.1); (ii) preregistered numerical campaign wording used consistently – NOT “experiment” (R35 s7.1); (iii) the RQL reconstruction + physical-parameter mapping; (iv) the DLGI state-space (Kalman/RTS) as ONE computational implementation only (R35 s7.2 – keep it out of the body); (v) ONE

FIG. 5 PLACEHOLDER (PASS-BRANCH) --- dual output. (a) comparator geometry: three columns (DCS-like dedicated monitor; pilot- interleaved GI; DLGI same raw record). (b) medium precision: forest/scatter of pilot-free/pilot RMSE ratios for t_c and CV across the confirmatory grid, 1.0 and 1.5 lines, oracle ratios light. (c) certification (MANDATORY): coverage vs cell for t_c and CV, $[0.92, 0.98]$ band, width-ratio inset. (d) scene + schedule: scene Δ PSNR vs byte-identical linear comparator, -0.2 dB line; paired/random/ordered ranking + reciprocity prediction. Narrow edge strip under b--d: fast/slow/weak-CV failure regions (NO sixth figure).

Figure 5: **[Fig. 5 title: One bucket record, two model-certified products.]** [Caption 120–160 words. Measures scientific outputs + uncertainty, not path correlation or smoother fit (R35 s7.2).]

pointer to the R19 analysis correction + immutable artifacts (register s9a); (vi) code/data availability. Full detail -i Supplement S1-S10. Materials-map pointers: paper-1 s2-s3 (main.oe.tex 162–239), paper2 s4 provenance, R34 campaign spec (materials map s4d).

Code and data availability

All simulation code, preregistration documents, manifests, correction records, and evidence bundles are available at [\[repo URL — USER DECISION \(gap G7\)\]](#).

Acknowledgments

[\[funding / acknowledgments — USER DECISION \(gap G7\)\]](#)